

A Beginner's Machine Learning Glossary

Artificial intelligence	A summary term for machines imitating human abilities
Machine learning	Using data to answer questions
Deep learning	Machine learning models with large, nested model architectures*
Garbage in, garbage out.	The understanding that if the data is low-quality, the model predictions will also be
Data science	A combination of coding skills, maths/stats expertise and domain knowledge
No free lunch - theorem	The best algorithm is dependent on the problem at hand
Supervised learning	Machine learning where the label of observations is known (ground truth*; e.g., classification)
Unsupervised learning	Machine learning where the label of observations is <i>not</i> known, pattern recognition (e.g., clustering)
Reinforcement learning	Machine learning models trained with a “reward function”, the model only learns by interacting with the training environment
Features / dimensions	The “independent variables”, the variables which we use to predict
Ground truth / gold standard	The “dependent variable”, the label or value we try to predict
Model	General term to describe the algorithm or the model after training
Algorithm	A list of steps (recipe) to achieve a certain task
Parameters	The values estimated during model fitting
Architecture	The structure of the machine learning model
Cost function / loss	The function that determines the goodness of the model fit
Task	The problem the machine learning model tries to solve
Epoch	How often a model cycles through all training data during model fitting
Batch	How many samples the model is presented at once during training
Training, validation, test set	The entire data is split into different parts to prevent over/underfitting* Training data: used to train the model Validation data: used to estimate model fit during training Test set: used to determine the final model performance after training
Overfitting / underfitting	The model adapts too well/too little to the training data
Tuning	Choosing or learning free parameters of the model
Fine-tuning	Re-training of a previously fitted model (e.g., to accomplish a related, but different task)
Pre-training	Training a model on a general purpose task
Transfer learning	Using existing models for related, but different tasks
CPU / GPU / TPU	Central processing unit: good for parallelization Graphical processing unit: good for matrix operations Tensor processing unit: a processor specialized on machine learning